

Understanding Flow Matching from a mathematical view

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Outline

- 1 Mathematical motivation for Flow Matching
- 2 Connection with Diffusion Models
- 3 Conditional Paths and Velocity Targets
- 4 Flow Matching Training and Inference
- 5 Advanced Topics and Extensions
- 6 MeanFlow: Single-Step Generation
- 7 Summary and Future Directions
- 8 References

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Probability Path and Coupling

- Let $\{p_t\}_{t \in [0,1]}$ be a prescribed path of marginals on $\mathbb{R}^d$ with p_0 simple and p_1 the data.
- A *coupling* induces random trajectories $t \mapsto x_t$ with instantaneous velocity $\dot{x}_t = \frac{d}{dt}x_t$ (may be stochastic given x_t).
- Goal: learn a *deterministic* vector field $v : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ such that the ODE

$$\frac{d}{dt}x_t = v(x_t, t), \quad x_0 \sim p_0,$$

yields marginals $x_t \sim p_t$ for all t .

Strong and Weak Continuity

From the Fokker–Planck equation, for $\frac{dx}{dt} = v(x, t)$, we have

$$\partial_t p_t(x) + \nabla \cdot (p_t(x) v(x, t)) = 0.$$

The corresponding weak form of above strong form is

Weak form (for any test functions $\varphi \in C_c^\infty(\mathbb{R}^d)$)

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) p_t(x) dx = \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v(x, t) p_t(x) dx,$$

i.e.,

$$\frac{d}{dt} \mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla \varphi(x_t) \cdot \dot{x}_t].$$

Weak Form from Random Trajectories

Let x_t be the coupled trajectory (not necessarily deterministic). For any test function φ ,

$$\frac{d}{dt} \mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla \varphi(x_t) \cdot \dot{x}_t].$$

Expanding the expectation into integrals at a fixed t :

$$\begin{aligned} \mathbb{E}[\nabla \varphi(x_t) \cdot \dot{x}_t] &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \nabla \varphi(x) \cdot \dot{x} \mu_t(dx, d\dot{x}) \\ &= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot \left(\int \dot{x} p(\dot{x} | x, t) d\dot{x} \right) p_t(x) dx \\ &= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v^*(x, t) p_t(x) dx = \mathbb{E}[\nabla \varphi(x_t) \cdot v^*(x, t)]. \end{aligned}$$

where $v^*(x, t) \equiv \mathbb{E}[\dot{x}_t | x_t = x, t] = \int_{\mathbb{R}^d} \dot{x} p(\dot{x} | x, t) d\dot{x}$. Thus the random path induces the weak identity with $v = v^*$.

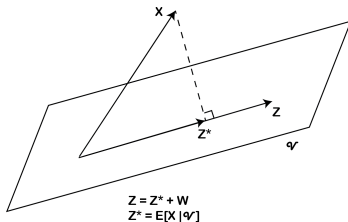
Theoretical Support: Conditional Expectation

Theorem

Let X be an integrable random variable. Then for each σ -algebra $\mathcal{V}$ and $Z \in \mathcal{V}$, $Z^* = \mathbb{E}(X|\mathcal{V})$ solves the least square problem

$$\operatorname{argmin}_{Z \in \mathcal{V}} \|Z - X\|,$$

where $\|Z\| = \left(\int Z^2 dP\right)^{\frac{1}{2}}$.



Hence we have $L = \mathbb{E}\|v_\theta(x_t, t) - \dot{x}_t\|^2$.

Practical Training Recipe (Per-Iteration)

- 1 Sample $t \sim w(t)$ (typically uniform on $[0, 1]$).
- 2 Sample $(x_t, \dot{x}_t)$ from the chosen path coupling.
- 3 Regress $\dot{x}_t$ on (x_t, t) via the MSE:

$$\mathcal{L} = \|v_\theta(x_t, t) - \dot{x}_t\|^2.$$

- 4 Update θ by SGD; no density estimation, no reverse simulation is needed during training.

Remark

Different path/coupling choices change v^ and the irreducible variance $\mathbb{E}\|\dot{x}_t - v^*\|^2$, affecting sample efficiency and optimization, but the optimal target remains the conditional mean.*

Key Takeaways

- The correct notion of “matching the path” is the **weak continuity** identity for all test functions.
- The only deterministic field depending on (x, t) that preserves this identity is the **conditional mean velocity**

$$v^*(x, t) = \mathbb{E}[\dot{x}_t \mid x_t = x, t] .$$

- The **flow-matching MSE** is precisely the L^2 projection selecting v^* :

$$v_\theta \xrightarrow[\text{minimize}]{\mathbb{E} \|v_\theta(x_t, t) - \dot{x}_t\|^2} v^* .$$

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Theorem (Fokker–Planck / Forward Kolmogorov)

Let $(X_t)_{t \geq 0} \in \mathbb{R}^N$ solve the Itô SDE

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t,$$

where $\mu : \mathbb{R}^N \times [0, \infty) \rightarrow \mathbb{R}^N$ is the drift, $\sigma : \mathbb{R}^N \times [0, \infty) \rightarrow \mathbb{R}^{N \times M}$ the diffusion, and W_t an M -dimensional standard Wiener process. Define the diffusion tensor

$$D(x, t) := \frac{1}{2} \sigma(x, t) \sigma(x, t)^\top \in \mathbb{R}^{N \times N}.$$

Then the (smooth) density $p(x, t)$ satisfies the Fokker–Planck equation

$$\partial_t p(x, t) = - \sum_{i=1}^N \frac{\partial}{\partial x_i} [\mu_i(x, t) p(x, t)] + \sum_{i,j=1}^N \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(x, t) p(x, t)]$$

Isotropic diffusion: $\sigma(x, t) = \sigma(t) \mathbf{I}_N$

When $\sigma(x, t) = \sigma(t) \mathbf{I}_N$, the diffusion tensor becomes

$$D(x, t) = \frac{1}{2} \sigma(t)^2 \mathbf{I}_N, \quad D_{ij}(x, t) = \frac{1}{2} \sigma(t)^2 \delta_{ij}.$$

Hence the Fokker–Planck equation simplifies to the drift–diffusion form

$$\partial_t p(x, t) = -\nabla \cdot (\mu(x, t) p(x, t)) + \frac{\sigma(t)^2}{2} \Delta p(x, t)$$

where $\nabla \cdot (\mu p) = \sum_{i=1}^N \partial_{x_i} (\mu_i p)$ and $\Delta = \sum_{i=1}^N \partial_{x_i}^2$ is the Laplacian.

Mathematical connection to diffusion models

Theoretical link: For a forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$, the probability flow ODE is:

$$\frac{d}{dt} \mathbf{x} = f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t). \quad (1)$$

Key mathematical insight: When Flow Matching uses Gaussian paths that match the SDE's marginal distributions p_t , the learned velocity field $\mathbf{v}_\theta$ approximates the probability flow drift:

$$\mathbf{v}_\theta(\mathbf{x}, t) \approx f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \quad (2)$$

Crucial advantage: Flow Matching achieves this approximation *without* ever computing the score $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training!

Practical implications: This connection allows Flow Matching to inherit the theoretical guarantees of diffusion models while avoiding their computational bottlenecks.

Mathematical derivation: probability flow ODE

Setup: Forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$ with density $p(\mathbf{x}, t)$ solving the Fokker-Planck equation:

$$\partial_t p = -\nabla \cdot (fp) + \frac{1}{2}g^2 \Delta p. \quad (3)$$

Key insight: A deterministic ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$ transports densities via the continuity equation:

$$\partial_t p + \nabla \cdot (p\mathbf{v}) = 0. \quad (4)$$

Mathematical derivation: Equating the two equations:

$$-\nabla \cdot (p\mathbf{v}) = -\nabla \cdot (fp) + \frac{1}{2}g^2 \Delta p \quad (5)$$

$$\Rightarrow \mathbf{v} = f - \frac{1}{2}g^2 \frac{\nabla p}{p} = f - \frac{1}{2}g^2 \nabla \log p. \quad (6)$$

Result: The probability flow ODE drift is $\mathbf{v} = f - \frac{1}{2}g^2 \nabla \log p$.

Connection to Flow Matching: When Flow Matching learns $\mathbf{v}_\theta$, it approximates this probability flow drift without computing the score $\nabla \log p$.

Questions

From

- ① Score model $s_\theta(x_t) \rightarrow s_{\theta^*}(x_t) \equiv \mathbb{E}[\nabla \log p(x_t|x_0) | x_t]$
- ② X prediction model (denoising model, x_0 model) $x_\theta(x_t) \rightarrow x_{\theta^*}(x_t) \equiv \mathbb{E}[x_0 | x_t]$;
- ③ ϵ prediction model $\epsilon_\theta(x_t) \rightarrow \epsilon_{\theta^*}(x_t) \equiv \mathbb{E}[\epsilon | x_t]$.

and

$$v^*(x_t) = \mathbb{E}[\dot{x}_t | x_t]$$

, what is the relationship between optimal $v_\theta(x, t)$ and the above models? Hints:

$$v_{\theta^*}(x_t) = \mathbb{E}[\dot{x}_t | x_t] = \mathbb{E}[\dot{\alpha}(t)x_t + \dot{\sigma}(t)\epsilon | x_t]. \quad (7)$$

and

$$s(x_t) = \mathbb{E}_x[\nabla_{x_t} \log p(x_t|x_0)|x_t] = \mathbb{E}_x[-\frac{x_t - \alpha x_0}{\sigma^2} | x_t] = -\frac{x_t - \alpha \mathbb{E}[x_0|x_t]}{\sigma^2}.$$

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- 8 References

Gaussian conditional paths: mathematical derivation

Assumption: Gaussian conditional paths

$$q_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t(\mathbf{x}_0), \sigma_t^2 \mathbf{I}), \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}. \quad (9)$$

Mathematical derivation: Taking the time derivative of the path:

$$\frac{d}{dt} \mathbf{x}_t = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} = \frac{\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t}. \quad (10)$$

Conditional expectation: Since $\boldsymbol{\epsilon}$ is independent of $\mathbf{x}_t$ given $\mathbf{x}_0$, we have:

$$\boxed{v = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0))}. \quad (11)$$

Mathematical analysis of path choices

1. Linear bridge (optimal transport):

- $\mu_t(\mathbf{x}_0) = (1 - t) \mathbf{x}_0, \sigma_t = t \sigma_{\max}$
- **Mathematical property:** Minimizes Wasserstein-2 distance
- **Velocity:** $v = \frac{\sigma_{\max}}{t}(\mathbf{x} - \mathbf{x}_0)$
- **Advantage:** Direct connection to optimal transport theory

2. VP-like (DDPM-inspired):

- $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0, \sigma_t = \sqrt{1 - \alpha^2(t)}$
- **Mathematical property:** Preserves energy structure
- **Velocity:** $v = \dot{\alpha}(t) \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \alpha(t) \mathbf{x}_0)$
- **Advantage:** Compatible with existing diffusion model schedules

3. VE-like (SMLD-inspired):

- $\mu_t(\mathbf{x}_0) = \mathbf{x}_0, \sigma_t = \sigma(t)$ increasing
- **Mathematical property:** Preserves data structure
- **Velocity:** $v = \frac{\dot{\sigma}(t)}{\sigma(t)}(\mathbf{x} - \mathbf{x}_0)$
- **Advantage:** Simple form, good for high-dimensional data

Key insight: All choices provide *closed-form* $\mathbf{u}_t$ via the boxed formula!

Derivation: VE/VP target forms

VE-like. $\mu_t(\mathbf{x}_0) = \mathbf{x}_0$, $\sigma_t = \sigma(t)$. Then

$$\mathbf{v} = 0 + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \mathbf{x}_0). \quad (12)$$

VP-like. $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0$, $\sigma_t = \sqrt{1 - \alpha^2(t)}$. Using $\mathbf{u}_t = \dot{\mu}_t + (\dot{\sigma}_t/\sigma_t)(\mathbf{x} - \mu_t)$, we get

$$\mathbf{v} = \dot{\alpha} \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} - \frac{\dot{\sigma}_t}{\sigma_t} \alpha \mathbf{x}_0 \quad (13)$$

$$= \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} + \left(\dot{\alpha} - \alpha \frac{\dot{\sigma}_t}{\sigma_t} \right) \mathbf{x}_0. \quad (14)$$

No explicit form for $\dot{\sigma}_t/\sigma_t$ is required to train; schedules determine it numerically.

Mathematical insight: These derivations show how different path choices lead to different velocity field structures, all computable analytically.

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Flow Matching: mathematical training objective

Theoretical foundation: Given conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0)$, the optimal velocity field satisfies:

$$\mathbf{v}^*(x, t) = \mathbb{E}[\dot{\mathbf{x}}_t \mid x_t = x, t] \quad (15)$$

Training procedure:

- 1 Sample $t \sim \mathcal{U}[0, 1]$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_t \sim q_t(\cdot \mid \mathbf{x}_0)$
- 2 Compute analytic target $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)$ from chosen path
- 3 Minimize regression loss

Mathematical objective:

$$J_{\text{FM}} = \mathbb{E}[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}\|^2]. \quad (16)$$

Key mathematical advantage: No score estimation needed; all targets are analytic!

Computational benefits:

- Stable gradients (no exploding/vanishing)
- Parallelizable across time steps
- No special architectural requirements

Mathematical foundation of inference

Theoretical basis: The learned velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ defines a deterministic ODE:

$$\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t), \quad t \in [0, 1] \quad (17)$$

Mathematical properties:

- **Existence:** Under Lipschitz conditions, solutions exist and are unique
- **Consistency:** The ODE preserves the continuity equation
- **Stability:** Deterministic integration avoids accumulation of stochastic errors

Generation process:

- 1 Initialize $\mathbf{x}(1) \sim p_1$ (e.g., $\mathcal{N}(0, \mathbf{I})$)
- 2 Integrate ODE backwards: $\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t)$, $t : 1 \rightarrow 0$
- 3 Use standard ODE solvers (Euler, RK4, Dormand-Prince, etc.)

Mathematical advantages:

- Deterministic sampling (reproducible)
- Fast with large solver steps
- No stochastic correctors needed

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- 8 References

Conditional Flow Matching

Extension to conditional generation: Instead of unconditional paths, we can define conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0, c)$ where c represents conditioning information (e.g., class labels, text prompts).

Mathematical formulation: The conditional velocity field becomes:

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0, c) := \mathbb{E}\left[\frac{d}{dt} \mathbf{x}_t \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0, c\right] \quad (18)$$

Training objective:

$$J_{\text{CFM}} = \mathbb{E}\left[\|\mathbf{v}_\theta(\mathbf{x}_t, t, c) - \mathbf{v}\|^2\right] \quad (19)$$

Applications:

- Class-conditional generation
- Text-to-image synthesis
- Image-to-image translation
- Style transfer

Advantage: Maintains the same computational efficiency as unconditional Flow Matching while enabling controlled generation.

Rectified Flow and Optimal Transport

Rectified Flow: A special case of Flow Matching that uses straight-line paths:

$$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1, \quad \mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0 \quad (20)$$

Mathematical properties:

- **Optimal transport:** Minimizes Wasserstein-2 distance
- **Simplicity:** Constant velocity field along each path
- **Efficiency:** Fastest convergence in many cases

Connection to optimal transport: Rectified Flow recovers the optimal transport map between p_0 and p_1 when the data is well-structured.

Practical advantages:

- Fewer function evaluations during sampling
- More stable training
- Better sample quality in many domains

Flow Matching in Latent Spaces

Motivation: Instead of working directly in pixel space, Flow Matching can be applied in learned latent representations.

Mathematical setup: Given an encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ and decoder $D : \mathcal{Z} \rightarrow \mathcal{X}$:

- Learn flow in latent space: $\mathbf{v}_\theta : \mathcal{Z} \times [0, 1] \rightarrow \mathcal{Z}$
- Generate latent codes: $\mathbf{z}_0 = \mathbf{z}_1 + \int_0^1 \mathbf{v}_\theta(\mathbf{z}_t, t) dt$
- Decode to images: $\mathbf{x} = D(\mathbf{z}_0)$

Advantages:

- **Computational efficiency:** Lower-dimensional space
- **Sample quality:** Better inductive biases in latent space
- **Flexibility:** Can use different architectures for encoder/decoder

Examples: Latent Diffusion Models (LDMs), Stable Diffusion

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MeanFlow: Motivation

Goal: Achieve single-step (1-NFE) generation while retaining high fidelity and stability.

Problem with standard Flow Matching: Multi-step ODE integration is computationally expensive and can accumulate errors.

Key insight: Learn a *ground-truth average velocity field* that summarizes an entire time interval.

Mathematical formulation: For any pair $r < t$ and point $\mathbf{z}_t$ along the flow path:

$$\bar{\mathbf{u}}(\mathbf{z}_t, r, t) := \frac{1}{t - r} \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau) \, d\tau. \quad (21)$$

Properties:

- $\bar{\mathbf{u}}$ is an **underlying field** induced by $\mathbf{v}$ (no networks yet)
- As $r \rightarrow t$, $\bar{\mathbf{u}}(\mathbf{z}_t, r, t) \rightarrow \mathbf{v}(\mathbf{z}_t, t)$
- Composition over intervals holds: one big step equals two smaller consecutive steps

Definition: Average Velocity

For any pair $r < t$ and point $\mathbf{z}_t$ along the flow path:

$$\bar{\mathbf{u}}(\mathbf{z}_t, r, t) := \frac{1}{t-r} \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau) \, d\tau. \quad (22)$$

- $\bar{\mathbf{u}}$ is an **underlying field** induced by $\mathbf{v}$ (no networks yet).
- As $r \rightarrow t$, $\bar{\mathbf{u}}(\mathbf{z}_t, r, t) \rightarrow \mathbf{v}(\mathbf{z}_t, t)$.
- Composition over intervals holds: one big step equals two smaller consecutive steps (additivity of integrals).

MeanFlow Identity (core relation)

Starting from Eq. (22):

$$(t - r) \bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau) d\tau. \quad (23)$$

Differentiate w.r.t. t (treating r as constant) and use the fundamental theorem of calculus:

$$\bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \mathbf{v}(\mathbf{z}_t, t) - (t - r) \frac{d}{dt} \bar{\mathbf{u}}(\mathbf{z}_t, r, t)$$

(24)

where the total derivative expands to

$$\frac{d}{dt} \bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \underbrace{\mathbf{v}(\mathbf{z}_t, t)}_{\frac{d\mathbf{z}_t}{dt}} \cdot \partial_{\mathbf{z}} \bar{\mathbf{u}}(\mathbf{z}_t, r, t) + \partial_t \bar{\mathbf{u}}(\mathbf{z}_t, r, t). \quad (25)$$

Interpretation: Eq. (24) links instantaneous and average velocities *without* introducing networks.

Mathematical significance: This identity enables learning average velocity fields that can be used for single-step generation.

Trainable objective

Parameterize a network $\bar{\mathbf{u}}_\theta(\mathbf{z}, r, t)$. Use the MeanFlow identity to form a target (replace $\mathbf{v}$ by sample-conditional $\mathbf{v}_t$):

$$\text{Target: } \bar{\mathbf{u}}_{\text{tgt}} = \mathbf{v}_t - (t - r) \underbrace{(\mathbf{v}_t \cdot \partial_{\mathbf{z}} \bar{\mathbf{u}}_\theta)}_{\text{JVP}} + \partial_t \bar{\mathbf{u}}_\theta. \quad (26)$$

$$\mathcal{L}(\theta) = \mathbb{E} \left\| \bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \text{sg}(\bar{\mathbf{u}}_{\text{tgt}}) \right\|_2^2. \quad (27)$$

Key components:

- $\text{sg}(\cdot)$: stop-gradient; avoids higher-order differentiation
- JVP (Jacobian-vector product) computed efficiently via autodiff
- If $r = t$, the correction term vanishes $\Rightarrow$ reduces to standard Flow Matching

Mathematical insight: This objective learns to predict average velocities over time intervals, enabling single-step generation.

Pseudocode (training)

MeanFlow training algorithm:

- 1 Sample $t, r \sim \text{LogitNormal}$; set $t \leftarrow \max(r, t)$, $r \leftarrow \min(r, t)$
- 2 Sample $\mathbf{x} \sim p_{\text{data}}$, $\epsilon \sim \mathcal{N}(0, I)$; set $\mathbf{z}_t = (1 - t)\mathbf{x} + t\epsilon$, $\mathbf{v}_t = \epsilon - \mathbf{x}$
- 3 Compute $(\bar{\mathbf{u}}_\theta, \frac{d}{dt}\bar{\mathbf{u}}_\theta) = \text{JVP}(\bar{\mathbf{u}}_\theta(\cdot, r, t), (\mathbf{v}_t, 0, 1))$
- 4 Set $\bar{\mathbf{u}}_{\text{tgt}} \leftarrow \mathbf{v}_t - (t - r) \frac{d}{dt}\bar{\mathbf{u}}_\theta$
- 5 Minimize $\|\bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \text{sg}(\bar{\mathbf{u}}_{\text{tgt}})\|_2^2$ (optionally with adaptive weights)

Key innovations:

- **LogitNormal sampling:** Ensures proper time interval distribution
- **JVP computation:** Efficient higher-order derivatives
- **Stop-gradient:** Prevents training instability

Sampling (1 step or few steps)

Replace the time integral by the learned average velocity:

$$\mathbf{z}_r = \mathbf{z}_t - (t - r) \bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t). \quad (28)$$

1-NFE sampling: take $(r, t) = (0, 1)$, $\mathbf{z}_1 = \epsilon$,

$$\boxed{\mathbf{x} \equiv \mathbf{z}_0 = \epsilon - \bar{\mathbf{u}}_\theta(\epsilon, 0, 1)} . \quad (29)$$

Mathematical significance: This single-step generation maintains the same theoretical guarantees as multi-step Flow Matching.

Practical advantages:

- **Speed:** Single function evaluation
- **Stability:** No accumulation of integration errors
- **Flexibility:** Can still use few-step variants by chaining intervals

CFG inside the field (still 1-NFE)

Define a CFG velocity field

$$\mathbf{v}_{\text{cfg}}(\mathbf{z}_t, t \mid c) = \omega \mathbf{v}(\mathbf{z}_t, t \mid c) + (1 - \omega) \mathbf{v}(\mathbf{z}_t, t). \quad (30)$$

Key insight: CFG can be applied directly to the average velocity field:

$$\bar{\mathbf{u}}_{\text{cfg}}(\mathbf{z}_t, r, t \mid c) = \omega \bar{\mathbf{u}}(\mathbf{z}_t, r, t \mid c) + (1 - \omega) \bar{\mathbf{u}}(\mathbf{z}_t, r, t). \quad (31)$$

Mathematical property: This maintains the 1-NFE property while enabling conditional generation.

Practical significance: MeanFlow can achieve both single-step generation and strong conditioning, combining the best of both worlds.

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- 7 Summary and Future Directions**
- 8 References

Key Takeaways: Flow Matching

Mathematical foundations:

- Flow Matching learns velocity fields that transport probability distributions via the continuity equation
- Conditional paths provide analytic supervision without score estimation
- The method connects to optimal transport theory and diffusion models

Computational advantages:

- No score estimation during training
- Deterministic sampling with flexible step sizes
- Stable gradients and efficient training

Practical benefits:

- Fewer function evaluations than diffusion models
- Better sample quality in many domains
- Natural extension to conditional generation

MeanFlow extension: Enables single-step generation while maintaining theoretical guarantees.

Future Directions and Open Problems

Theoretical developments:

- **Path optimization:** Finding optimal conditional paths for specific data distributions
- **Convergence analysis:** Understanding when Flow Matching achieves optimal transport
- **Error bounds:** Quantifying approximation errors in finite-sample settings

Practical improvements:

- **Architecture design:** Specialized networks for different data modalities
- **Training strategies:** Adaptive loss weighting and curriculum learning
- **Sampling efficiency:** Better ODE solvers and step size selection

Applications:

- **Scientific computing:** Molecular dynamics, fluid simulation
- **Creative AI:** Music, video, and 3D content generation
- **Data augmentation:** Synthetic data for training other models

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- 6 MeanFlow: Single-Step Generation
- 7 Summary and Future Directions
- 8 References**

Key References

Core Flow Matching papers:

- Lipman et al., *Flow Matching: A unifying perspective of score-based training and generative modeling* (2023)
- Liu et al., *Flow Matching for Generative Modeling* (2023)
- Albergo et al., *Building Normalizing Flows with Stochastic Interpolants* (2023)

MeanFlow and extensions:

- Song et al., *MeanFlow: A Flow Matching Method for One-Step Generation* (2024)
- Song et al., *Rectified Flow: A Marginal Preserving Approach to Optimal Transport* (2022)

Related work:

- Song et al., *Score-based generative modeling through stochastic differential equations* (2021)
- Ho et al., *Denoising Diffusion Probabilistic Models* (2020)
- Chen et al., *Neural Ordinary Differential Equations* (2018)

Implementation resources:

- Flow Matching Guide and Code (GitHub)
- Open source libraries: diffusers, torchdyn

Welcome inputs

Any comments?

Welcome your inputs!